University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

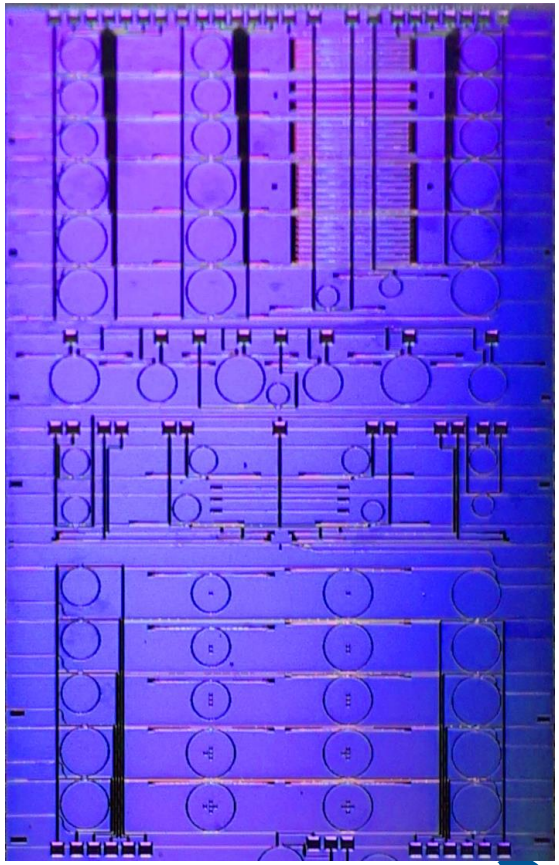
Lecture 15



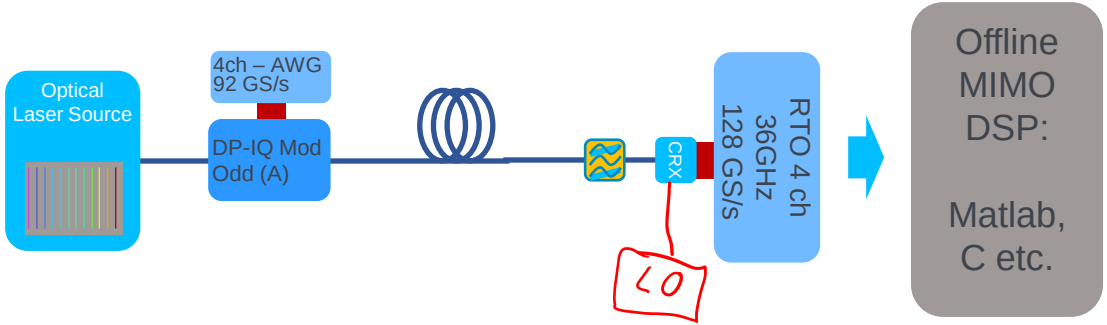
Prof. Dr.-Ing. Georg Rademacher

Photonic integrated systems – new course in SoSe 2026!

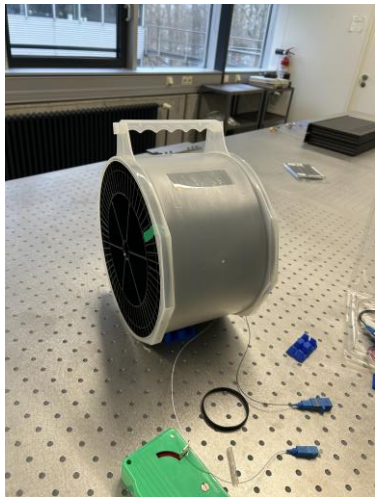
- Foundations of Integrated Photonics — Motivation, applications, material platforms, and optical waveguide fundamentals including modes, confinement, loss, and dispersion.
- Core Photonic Components — Passive building blocks (couplers, splitters, MMIs), interferometric structures (Mach–Zehnder), and resonant devices (ring resonators) for routing and filtering light on chip.
- Active Functionality & Integration — Modulators, detectors, and sources, along with platform tradeoffs, fabrication constraints, and design workflows using foundry processes.
- Circuit-Level Perspective & Applications — Integration of devices into photonic circuits and systems, case studies from communications, sensing, or quantum photonics, and emerging research trends.



Transmission demonstrations – Experimental setup



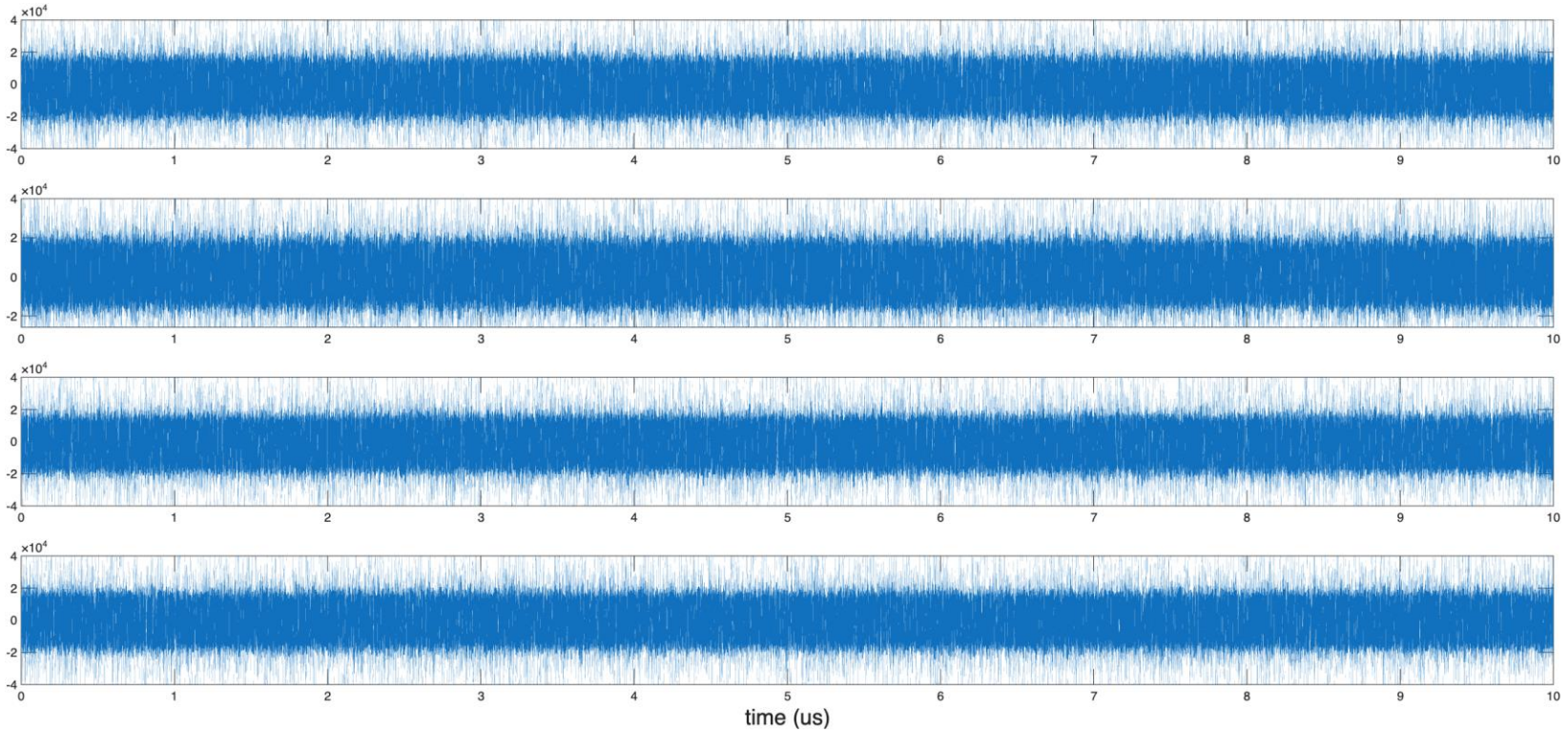






Traces from the Scope

University of Stuttgart
Germany



Digital Signal Processing (DSP)

1164 IEEE JOURNAL OF SELECTED TOPICS IN QUANTUM ELECTRONICS, VOL. 16, NO. 5, SEPTEMBER/OCTOBER 2010

Digital Coherent Optical Receivers: Algorithms and Subsystems

Seb J. Savory, *Member, IEEE*

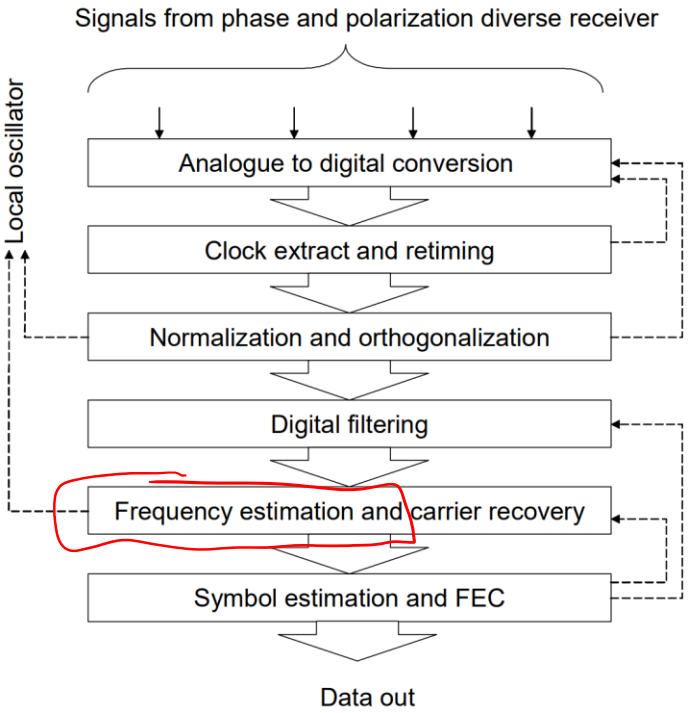
(Invited Paper)

***Abstract*—Digital coherent receivers have caused a revolution in the design of optical transmission systems, due to the subsystems and algorithms embedded within such a receiver. After giving a high-level overview of the subsystems, the optical front end, the analog-to-digital converter (ADC) and the digital signal processing (DSP) algorithms, which relax the tolerances on these subsystems are discussed. Attention is then turned to the compensation of transmission impairments, both static and dynamic. The discussion of dynamic-channel equalization, which forms a significant part of the paper, includes a theoretical analysis of the dual-polarization constant modulus algorithm, where the control surfaces several different equalizer algorithms are derived, including the constant modulus, decision-directed, trained, and the radially directed equalizer for both polarization division multiplexed quadriphase shift keyed (PDM-QPSK) and 16 level quadrature am-**

maximum data per wavelength (5.1 Tb/s) [9], and the maximum capacity distance product (112 Pbkm/s) [10], coherent detection also offers benefits at the network level. Not only can digital coherent receivers allow for more robust transmission, and hence, rerouting, but due to the presence of a local oscillator, the receiver is inherently frequency selective permitting new network architectures to be devised [11], [12].

In order to understand the operation of the digital coherent receiver, in this paper, we consider each of the subsystems required. After giving a high-level overview of the subsystems in Section II, we discuss the optical front end, analog-to-digital converter (ADC) and the DSP algorithms necessary to overcome

Digital Signal Processing (DSP)



$$r(n) = s(n)\exp(j(2\pi \Delta f_n \cdot T + \phi)) + N(n)$$

$s(n)$: $\pm 1 \pm j$

Δf : frequency offset

$$s(n)^4 = 1$$

$$r(n)^4 = s(n)^4 \cdot \exp(4 \cdot j(2\pi \Delta f_n T + \phi))$$

662 JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 28, NO. 4, FEBRUARY 15, 2010

Capacity Limits of Optical Fiber Networks

René-Jean Essiambre, *Senior Member, IEEE, Fellow, OSA*, Gerhard Kramer, *Fellow, IEEE*,
Peter J. Winzer, *Fellow, IEEE*, Gerard J. Foschini, *Fellow, IEEE*, and Bernhard Goebel, *Student Member, IEEE*

(Invited Paper)

***Abstract*—We describe a method to estimate the capacity limit of fiber-optic communication systems (or "fiber channels") based on information theory. This paper is divided into two parts. Part 1 reviews fundamental concepts of digital communications and information theory. We treat digitization and modulation followed by information theory for channels both without and with memory. We provide explicit relationships between the commonly used signal-to-noise ratio and the optical signal-to-noise ratio. We further evaluate the performance of modulation constellations such as quadrature amplitude modulation, combinations of am-**

and practical interests [1]–[4]. Such a limit is referred to as the *channel capacity* and the process of evaluating this limit leads to a better understanding of the technologies needed to approach it.

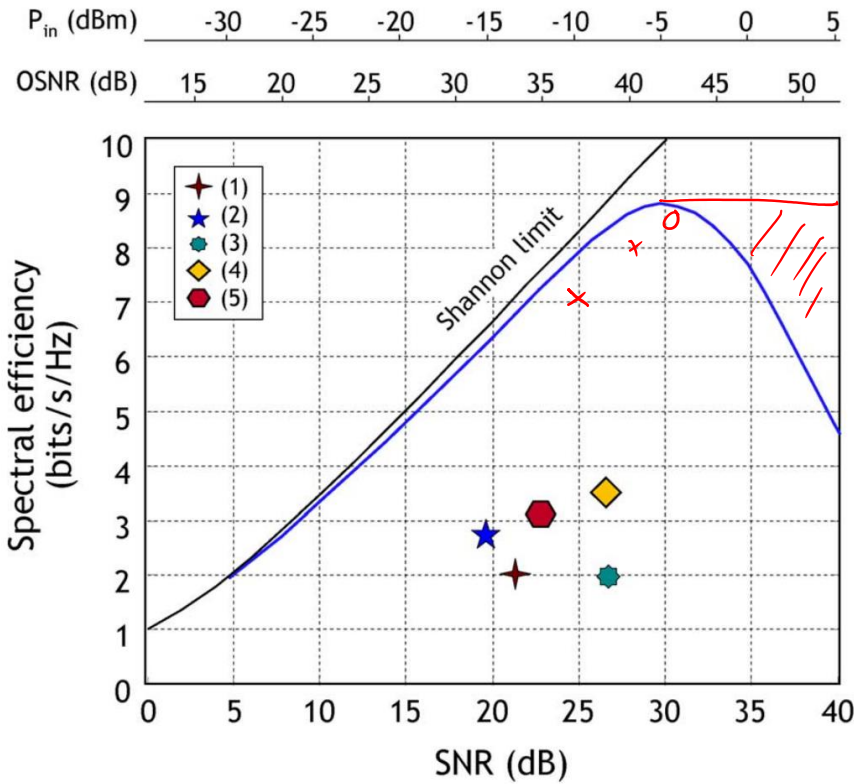
Capacity evaluations require information theory [1] that must be adapted to the specific characteristics of the channel under study. A pragmatic approach is to define the channel as "that part of a communication system that the designer is unable or un-

Capacity Limits of Optical Communications systems

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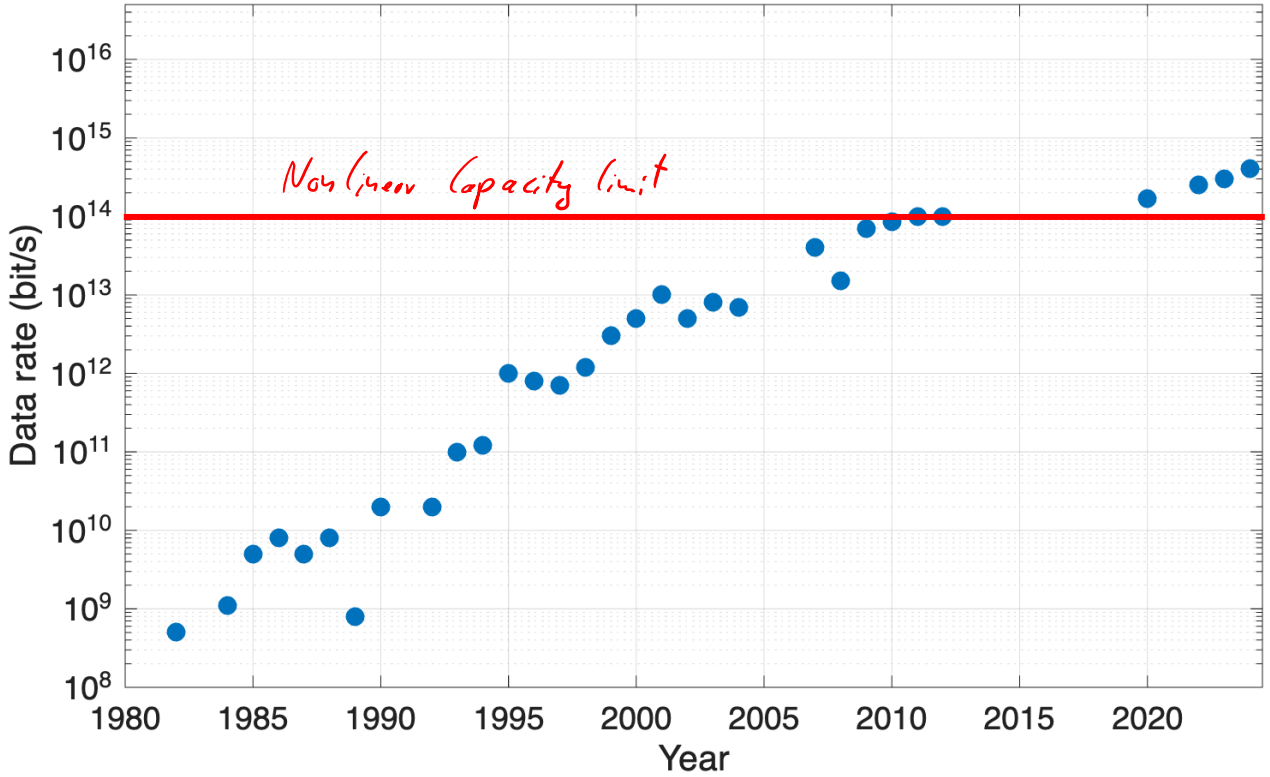
$$SNR_{NL} = \frac{P_{CH}}{P_{ASE} + \eta P_{CH}^3}$$

$$SE = \log_2(1 + SNR)$$

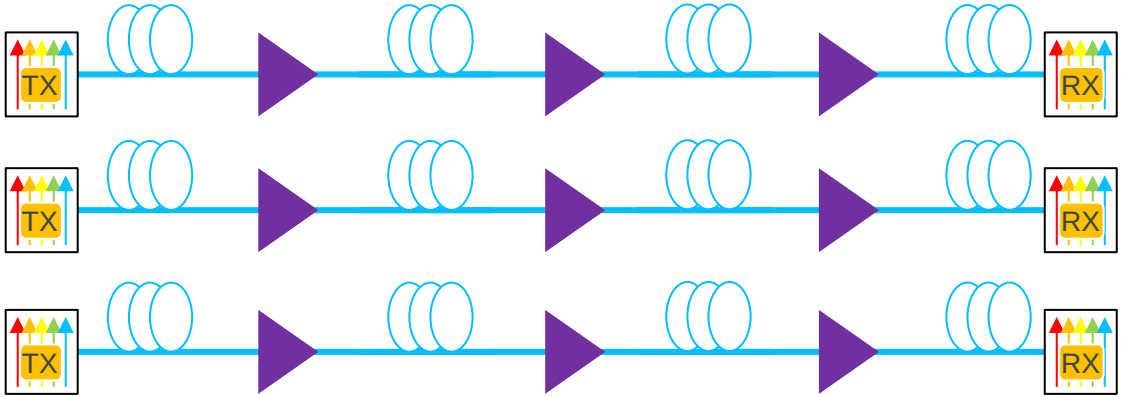


Space Division Multiplexed Transmission

University of Stuttgart
Germany

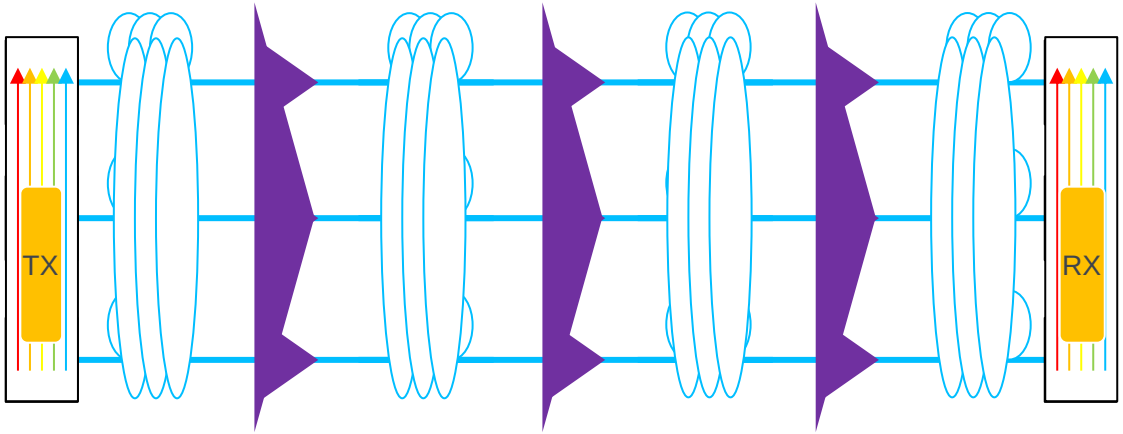


Space-division multiplexing (SDM)



Space-division multiplexing (SDM)

University of Stuttgart
Germany

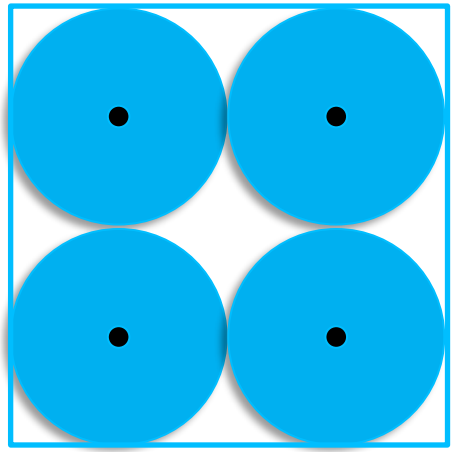


Optical fibers for space-division multiplexing

University of Stuttgart
Germany

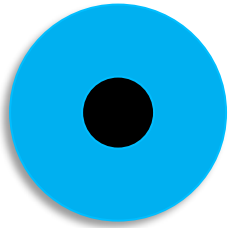
Fiber bundle

$\sim 125\ \mu m$



Few-/ multi-mode fiber

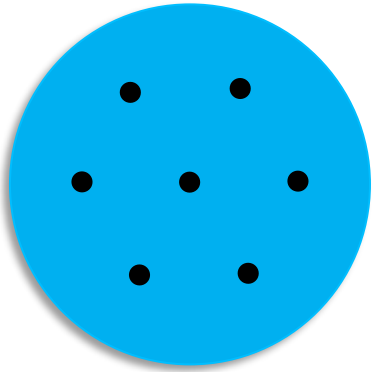
$\sim 125\ \mu m$



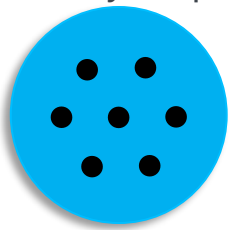
Multi-core fiber

$\sim 125\ -\ 250\ \mu m$

Weakly coupled



Randomly coupled



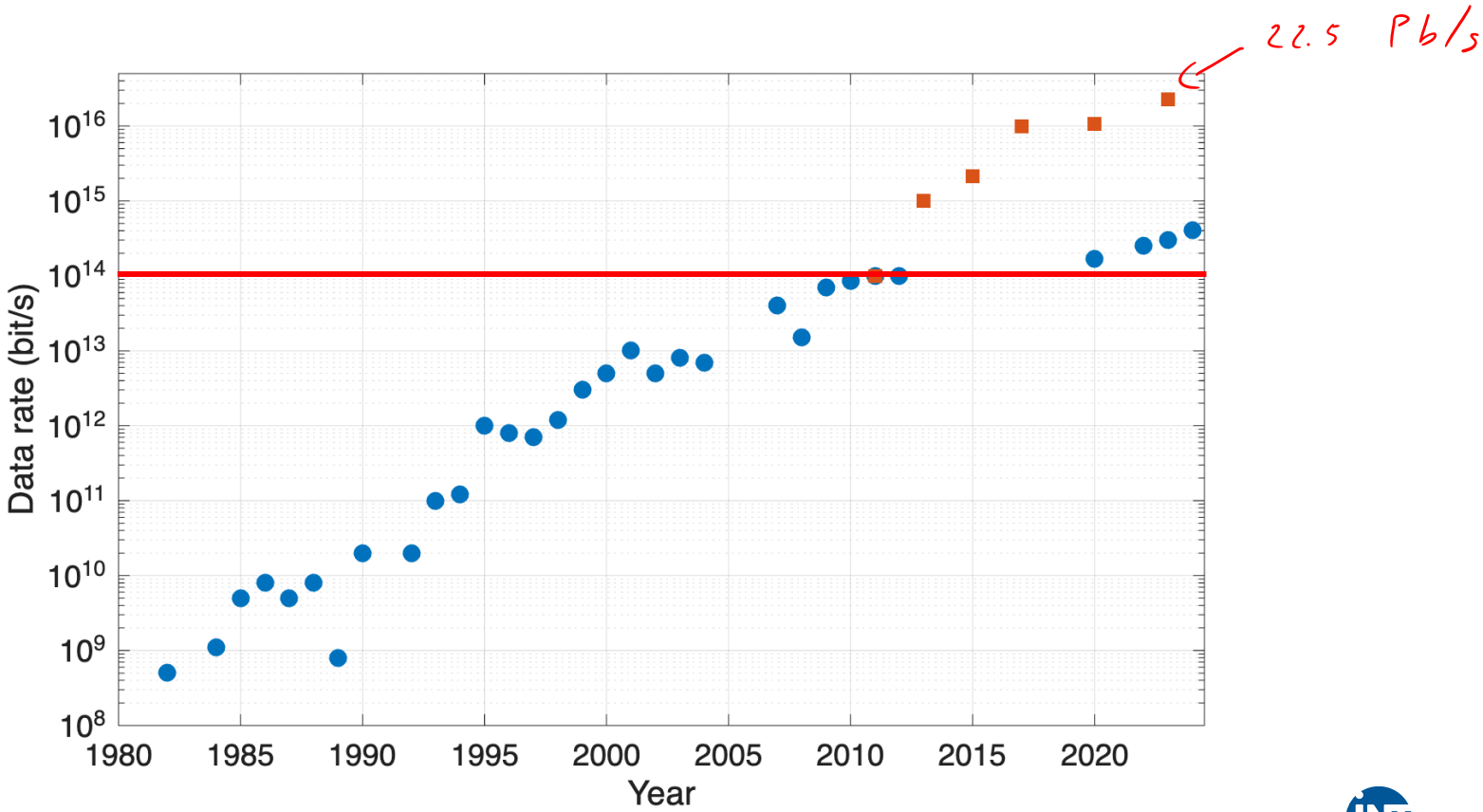
Few-mode-multi-core fiber

$\sim 160\ -\ 320\ \mu m$



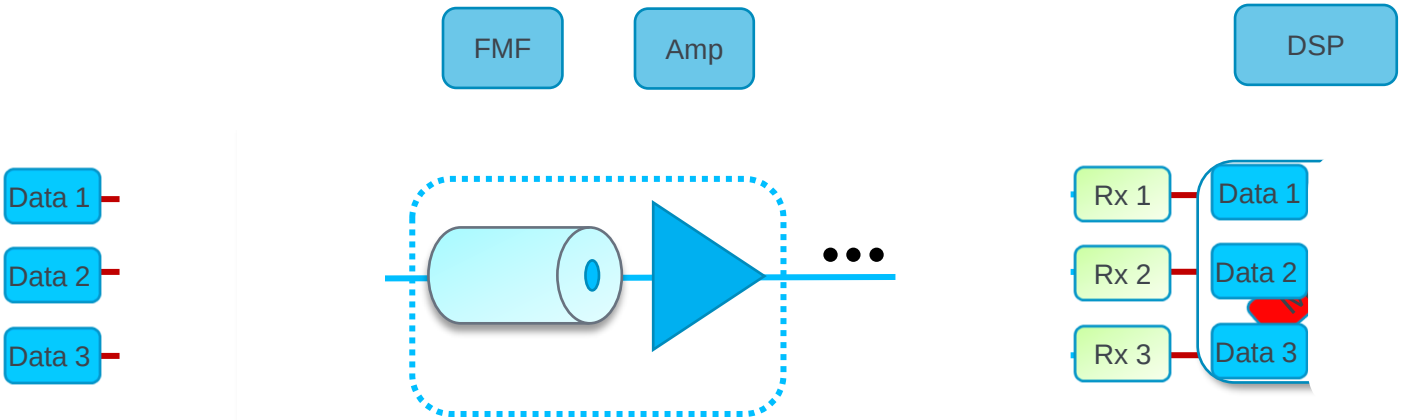
Space Division Multiplexed Transmission

University of Stuttgart
Germany



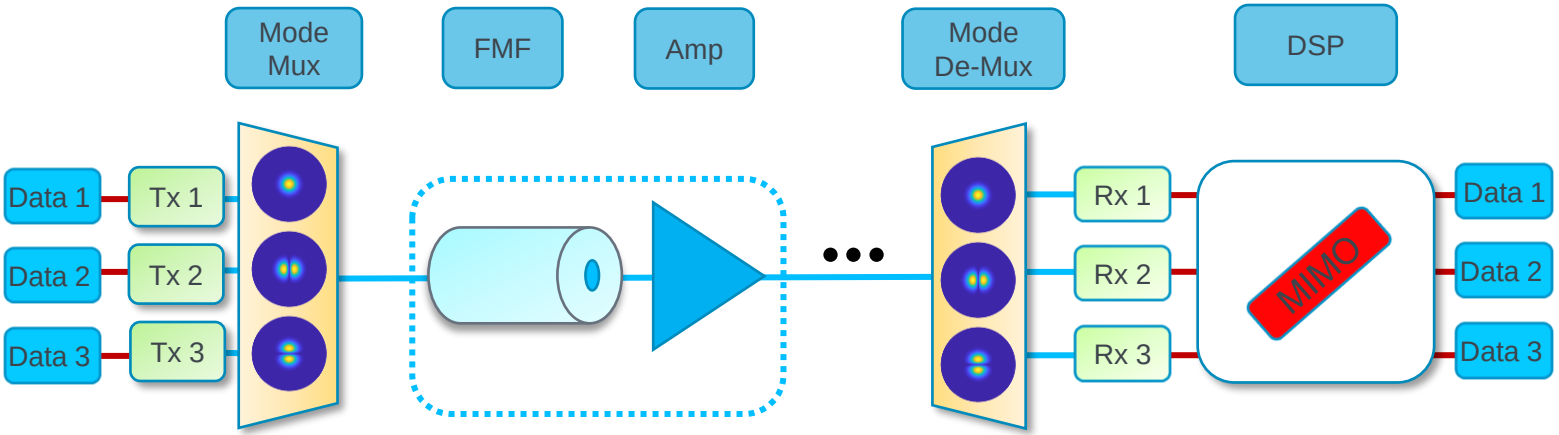
SDM transmission with a three-mode fiber

University of Stuttgart
Germany



SDM transmission with a three-mode fiber

University of Stuttgart
Germany



Output Channel matrix Input

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix}^{-1} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

Capacity of an SDM system

University of Stuttgart
Germany

MIMO capacities and outage probabilities in spatially multiplexed optical transport systems

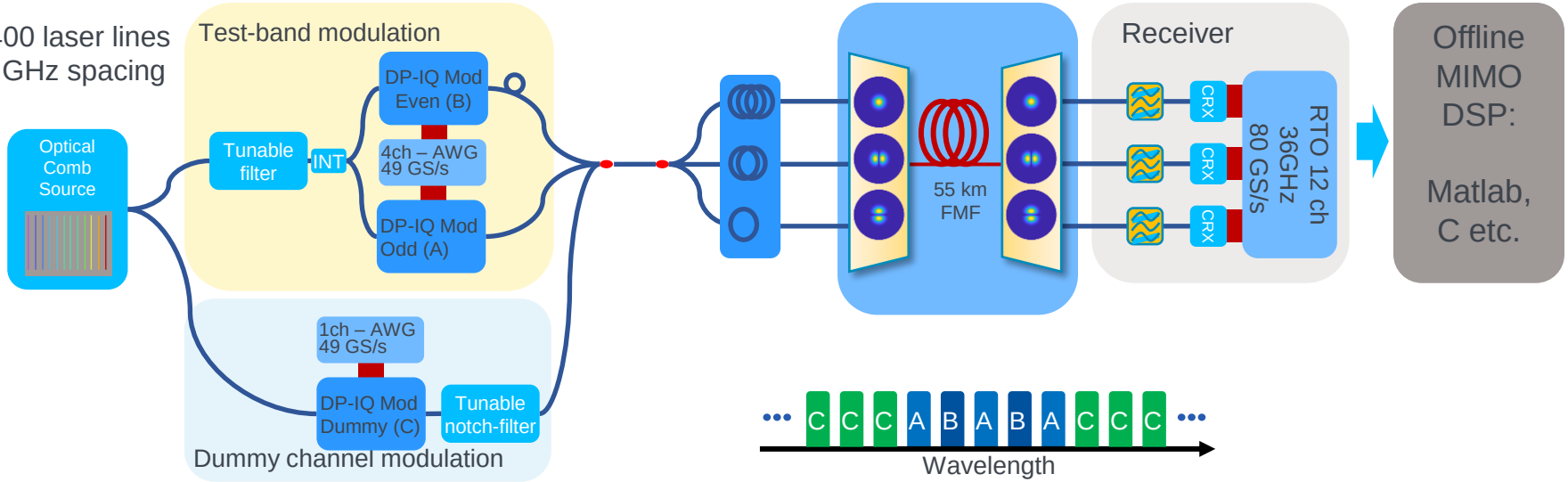
Peter J. Winzer* and Gerard J. Foschini
Bell Labs, Alcatel-Lucent, 791 Holmdel-Keyport Rd., Holmdel, New Jersey 07733, USA
**peter.winzer@alcatel-lucent.com*

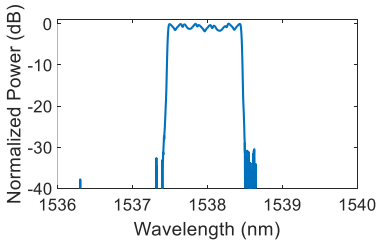
$$C = \sum_{i=1}^{r} \log_2\left(1 + \lambda_i \frac{E_s}{N_0}\right)$$

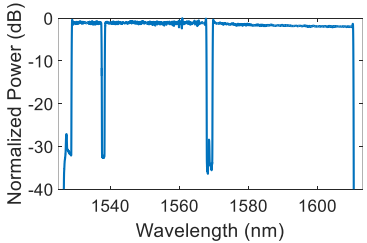
λ_i : Eigenvalues $H \cdot H^+$

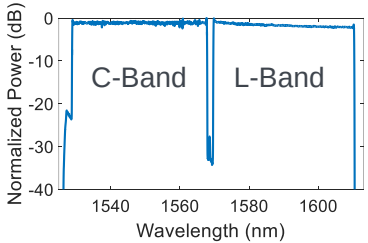
Transmission demonstrations – Experimental setup

- > 400 laser lines
- 25 GHz spacing



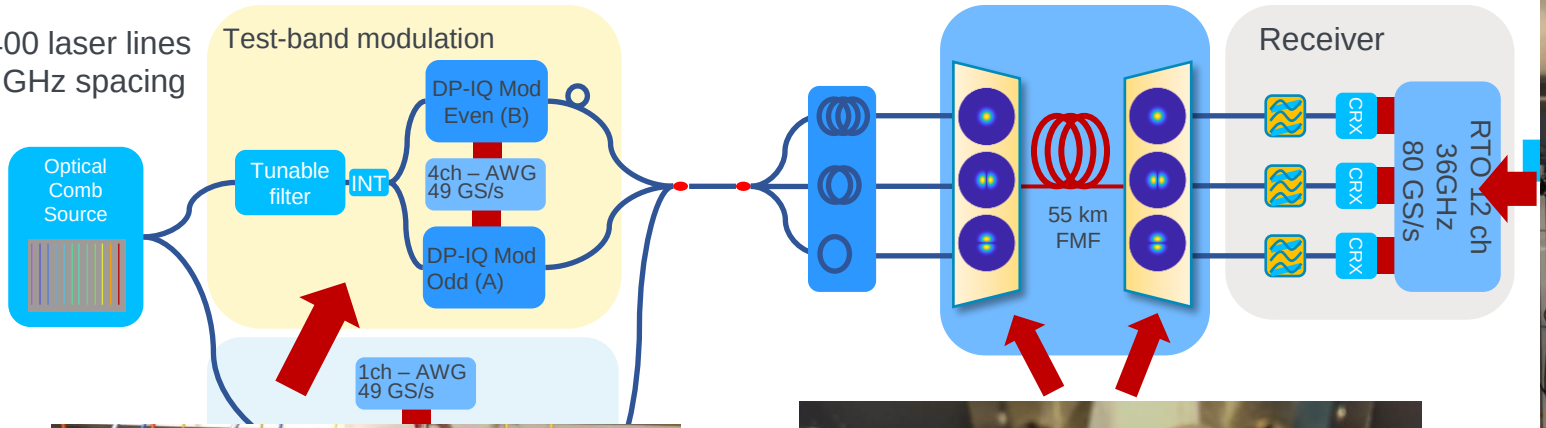


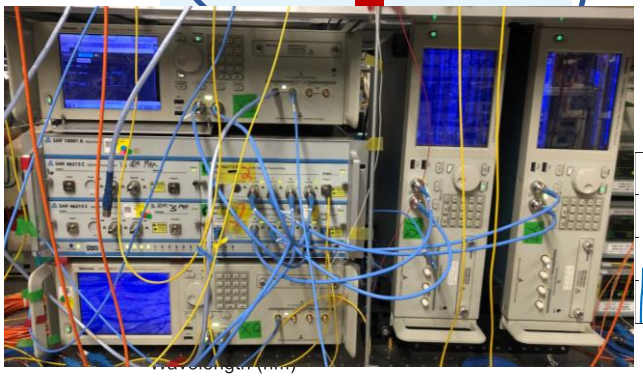


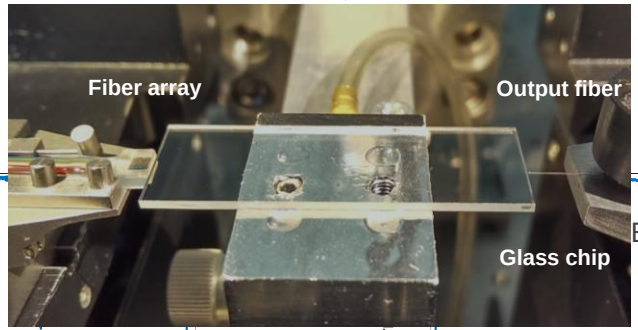


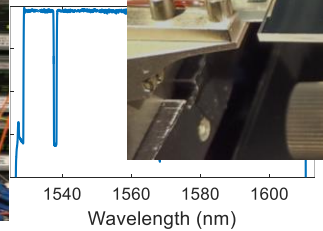
Transmission demonstrations – Experimental setup

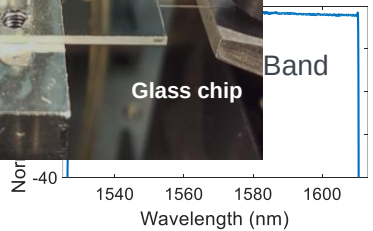
- > 400 laser lines
- 25 GHz spacing







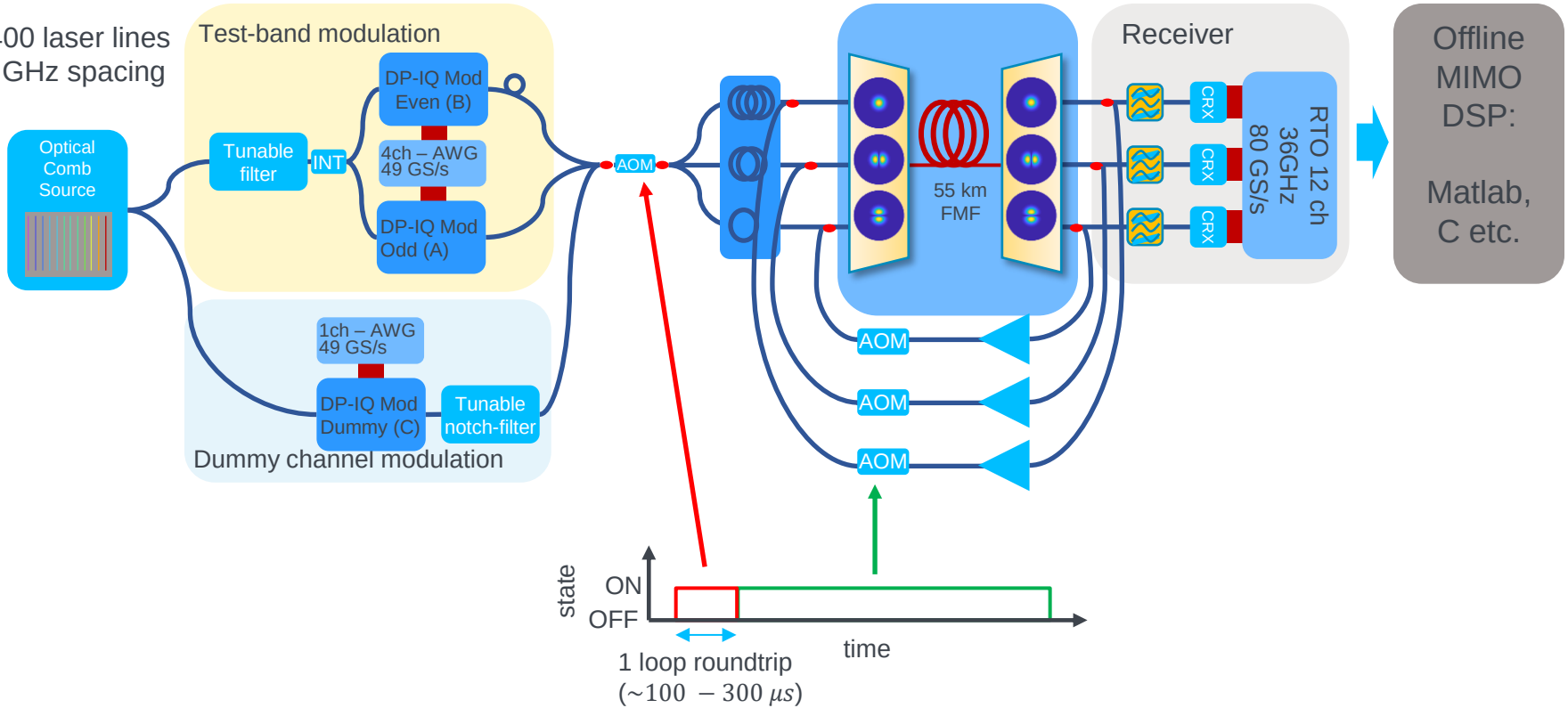




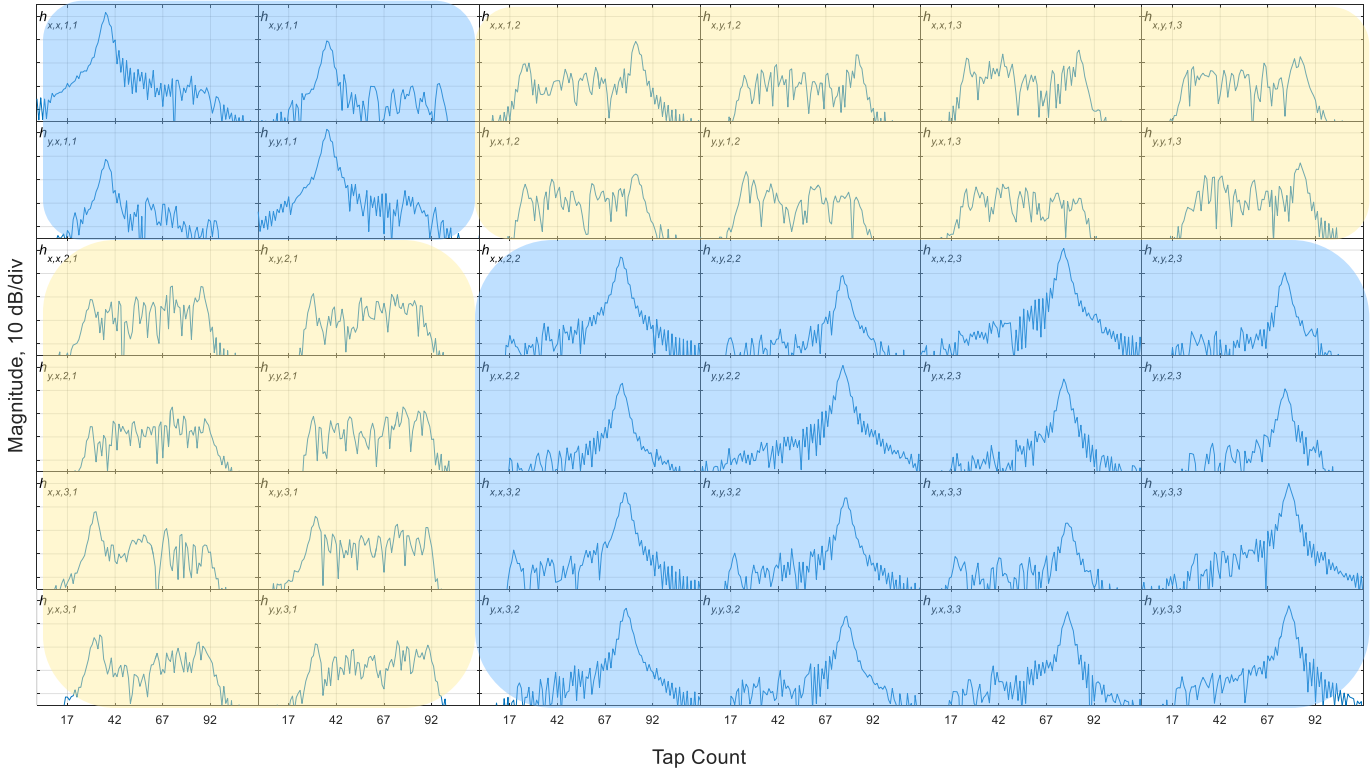


Transmission demonstrations – Experimental setup

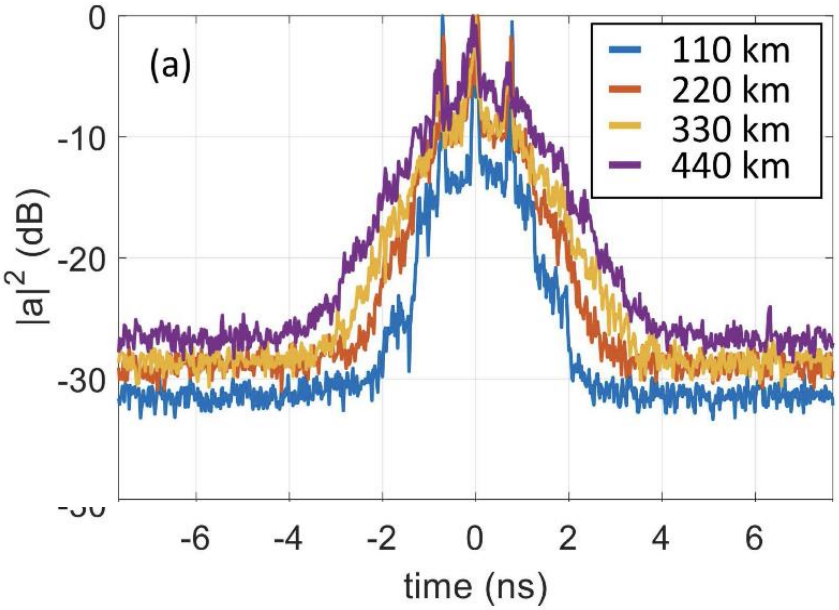
- > 400 laser lines
- 25 GHz spacing

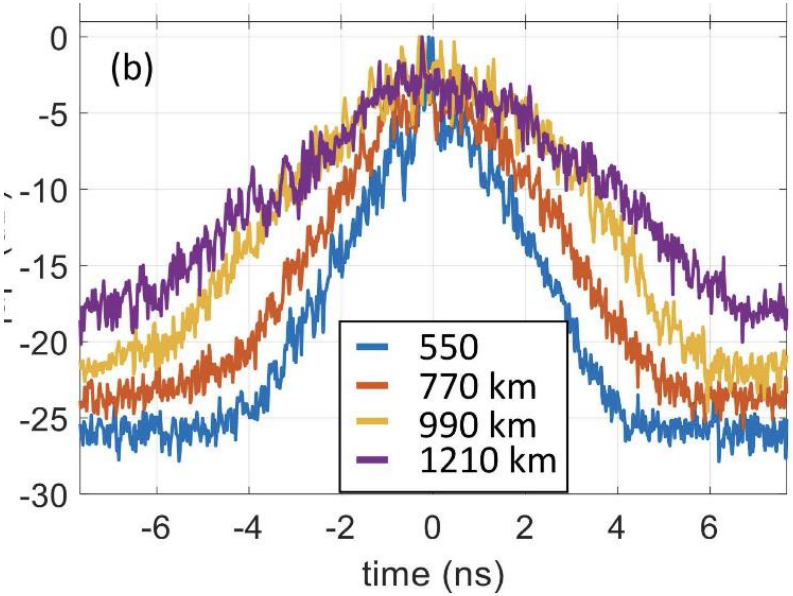


Understanding the transmission channel



Understanding the transmission channel: DMD-spread

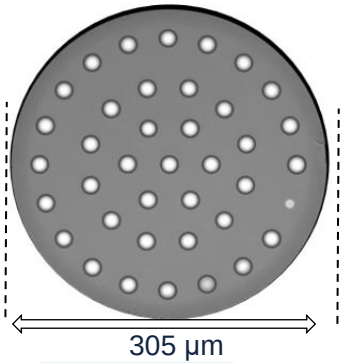




Record SDM fiber demonstrations

University of Stuttgart
Germany

13 km 38-few-mode multi-core fiber (MCF)

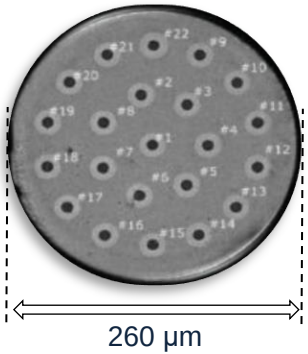


305 µm

2020
10.66 Pb/s
(C+L band)

2023
22 Pb/s
(S/C/L band)

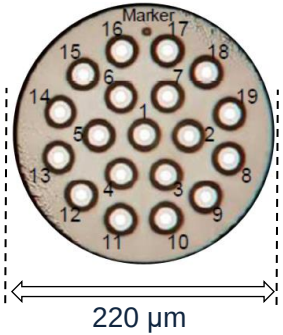
31.4 km 22-core MCF



260 µm

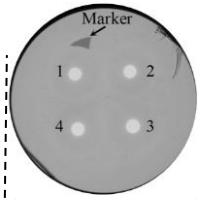
2015
2.15 Pb/s

30 km 19-core MCF



220 µm

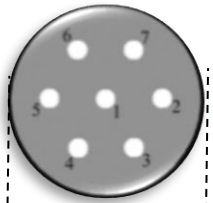
3.5 km 4-core few-mode MCF.



160 µm

2018
1.2 Pb/s

53.7 km 7-core MCF



160 µm

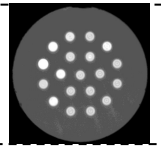
54-100 km 4-core MCF



125 µm

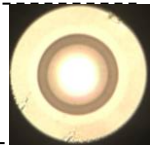
2022
1.02 Pb/s

66 km randomly coupled MCF



2023
1.7 Pb/s

26 km 55-mode fiber



2023
3.56 Pb/s

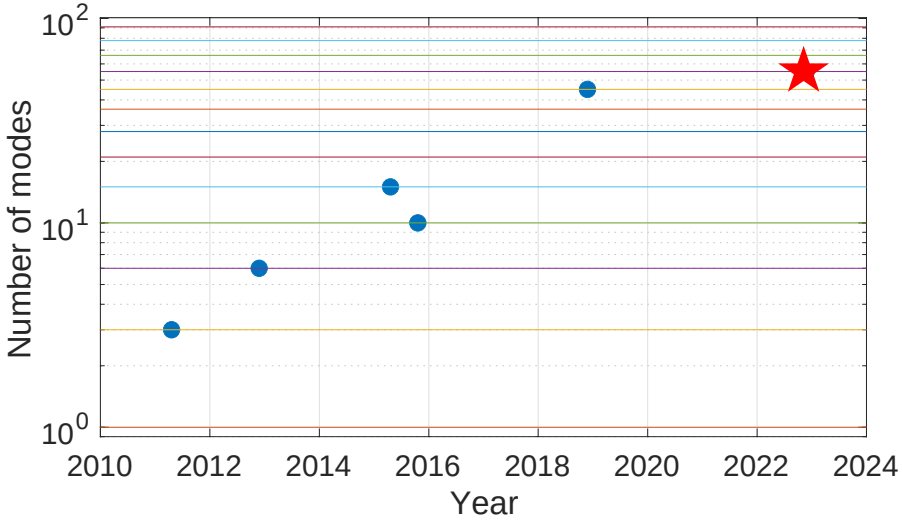
3.56 Peta-bit/s C+L Band Transmission over a 55-mode Multi-Mode Fiber

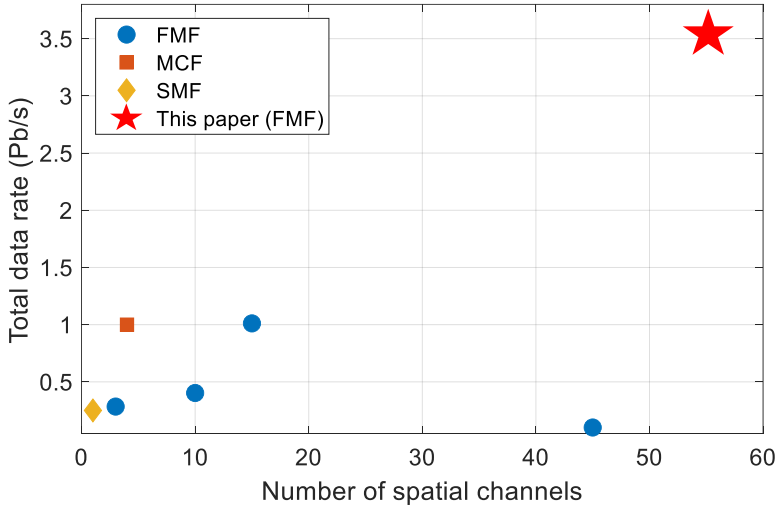
Georg Rademacher, Ruben S. Luís, Ben Puttnam, Nicolas K. Fontaine, Mikael Mazur, Haoshuo Chen, Roland Ryf, David T. Neilson, Daniel Dahl, Joel Carpenter, Pierre Sillard, Frank Achten, Marianne Bigot, Jun Sakaguchi, and Hideaki Furukawa

European Conference on Optical Communication (ECOC), Top-scored paper, 2023

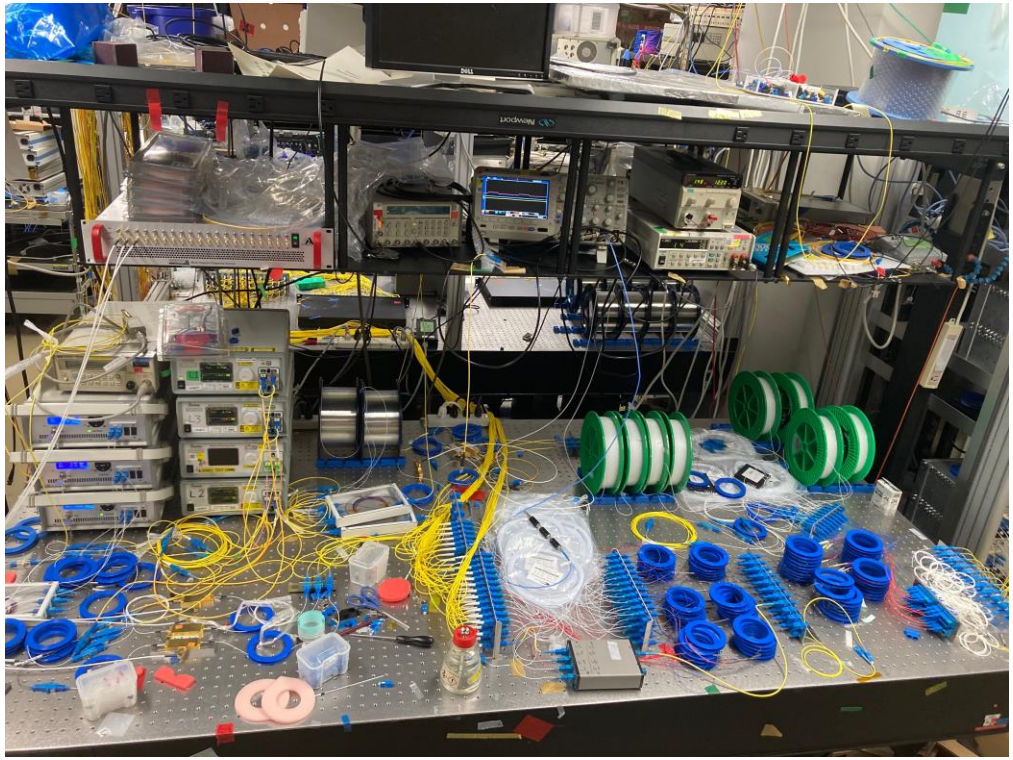
Motivation for 55 mode transmission

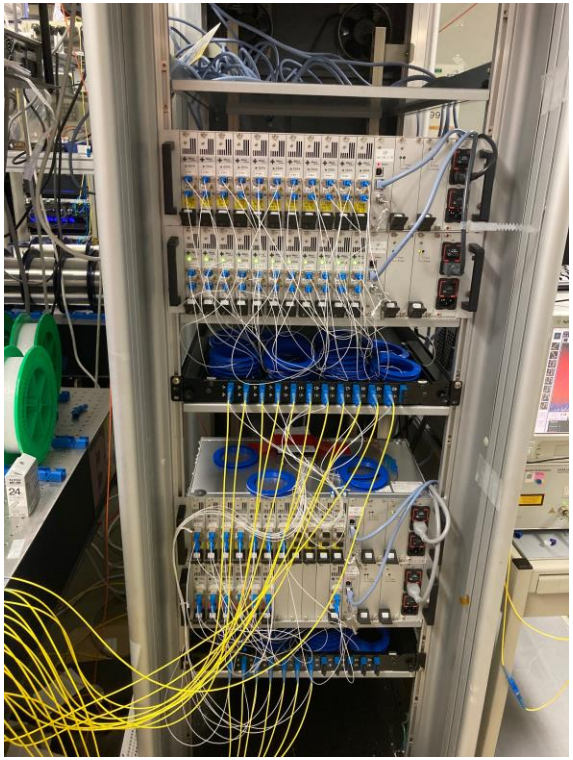
University of Stuttgart
Germany





Snapshot of lab system





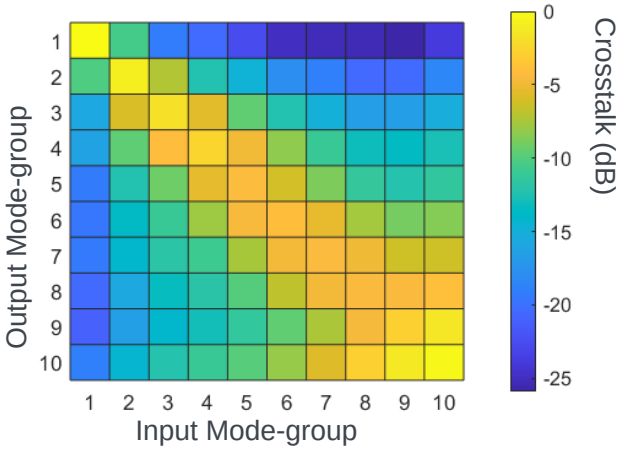
Snapshot of lab system

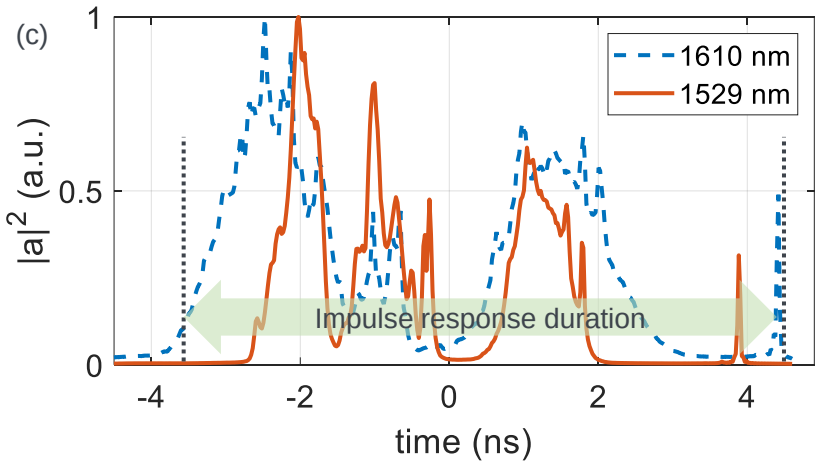


Snapshot of lab system



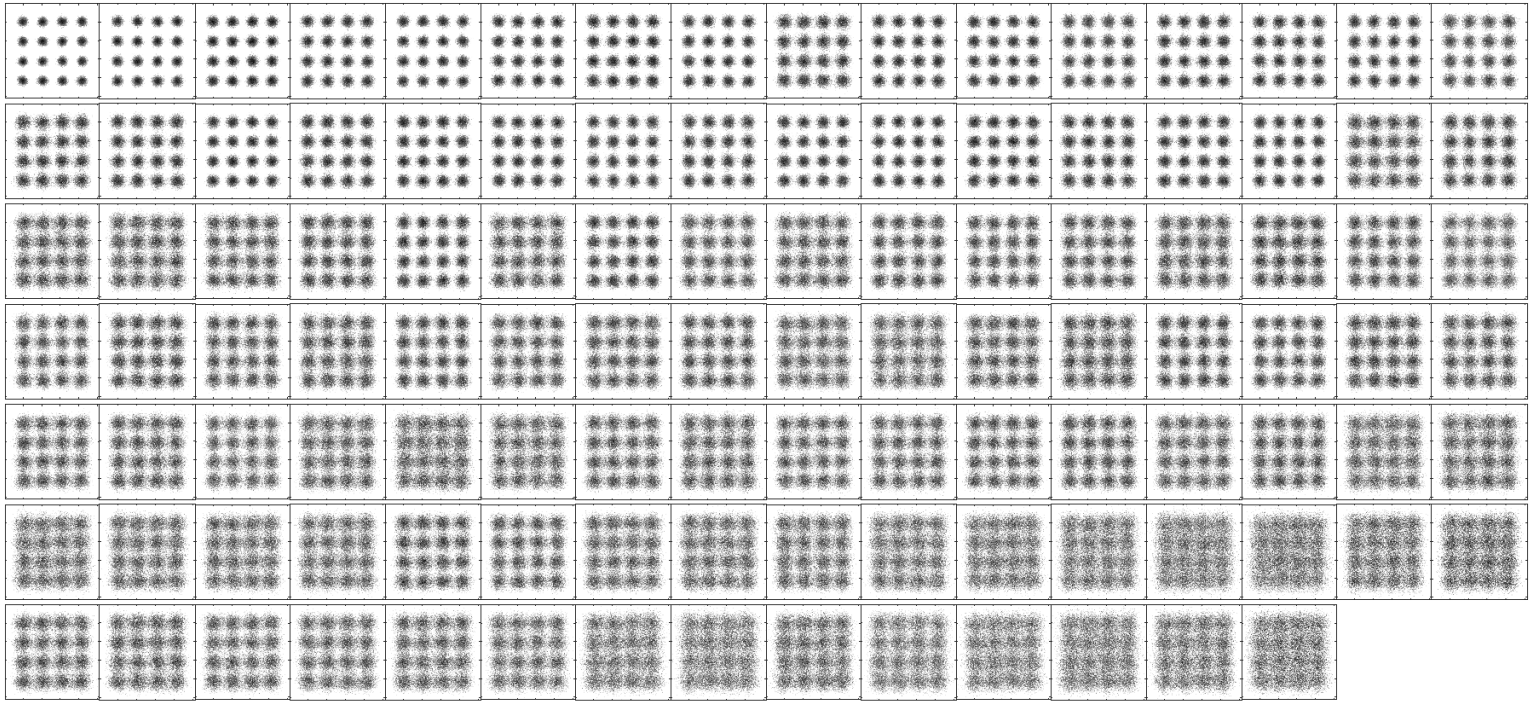
Transmission channel characteristics:



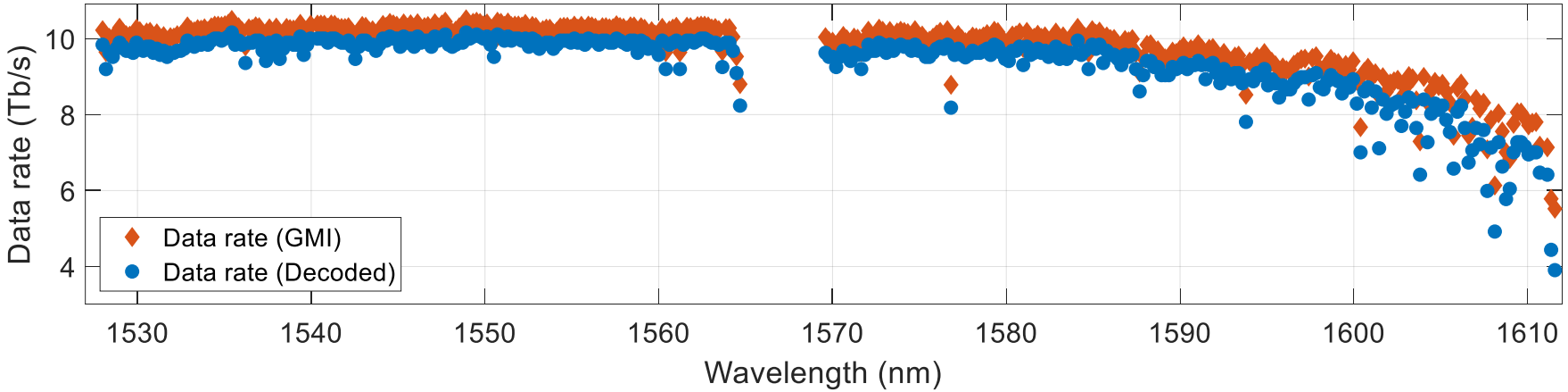


Constellations of 55 modes x 2 polarizations

Wavelength channel at 1543 nm



Data rates of 383 wavelength channels

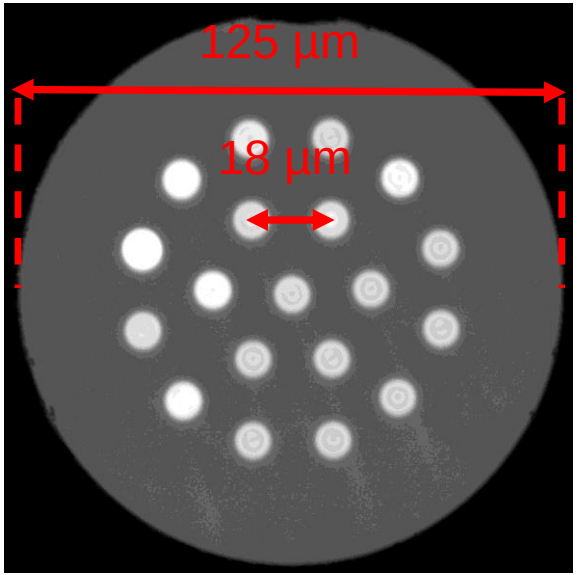


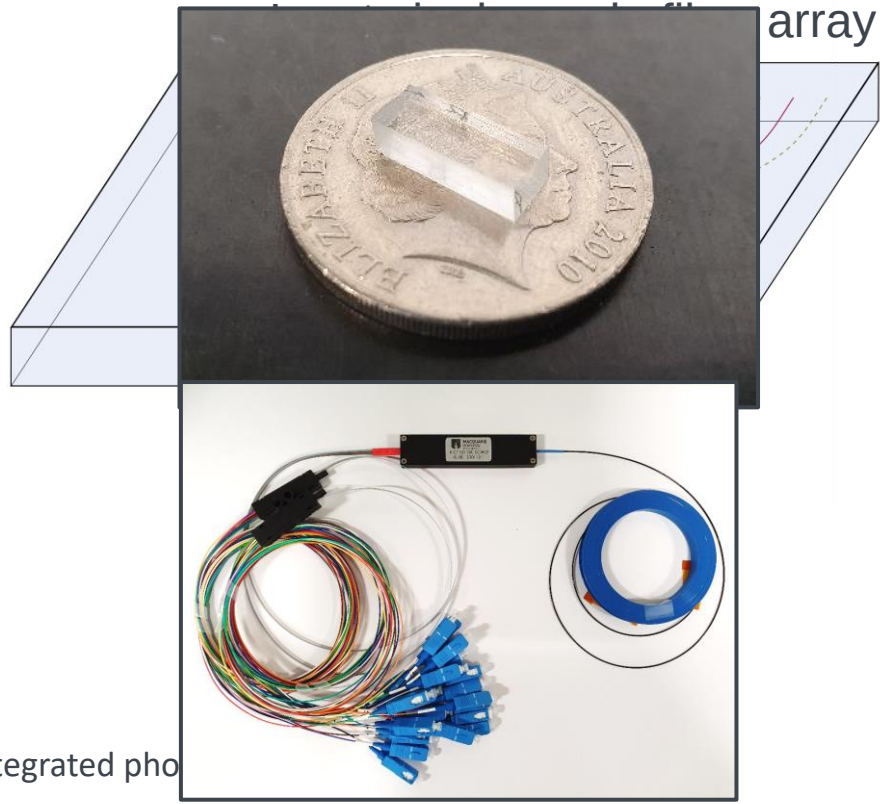
Randomly Coupled 19-Core Multi-Core Fiber with Standard Cladding Diameter

Georg Rademacher, Menno van den Hout, Ruben S. Luís, Ben Puttnam, Giammarco Di Sciullo, Tetsuya Hayashi, Ayumi Inoue, Takuji Nagashima, Simon Gross, Andrew Ross-Adams, Michael J. Withford, Cristian Antonelli, Chigo Okonkwo, Jun Sakaguchi, and Hideaki Furukawa

Optical fiber communications conference (OFC), Post-Deadline paper, 2023

Fiber Design and Core Multiplexer



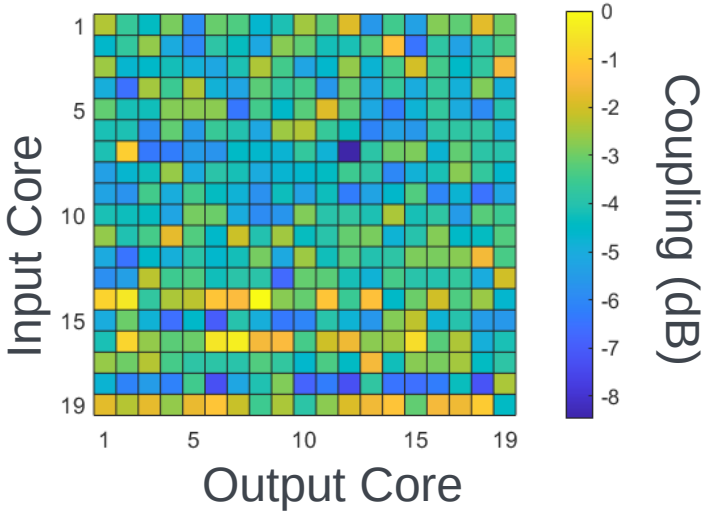


- Pure silica cores
- Effective area 62 µm^2
- 0.215 dB/km attenuation at 1550 nm

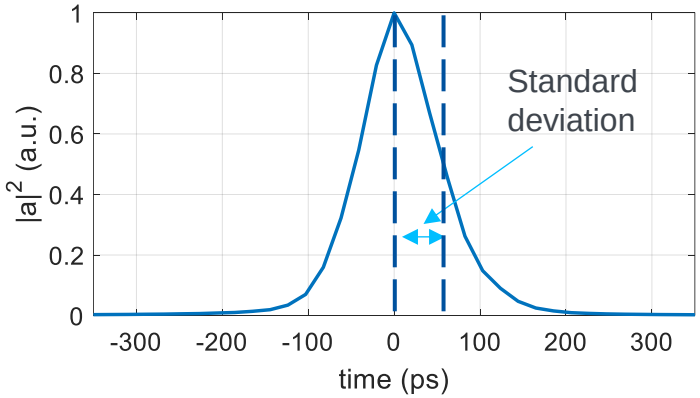
S. Gross and M.J. Withford, ”Ultra fast-laser-inscribed 3D integrated pho

Transmission Channel Characteristics

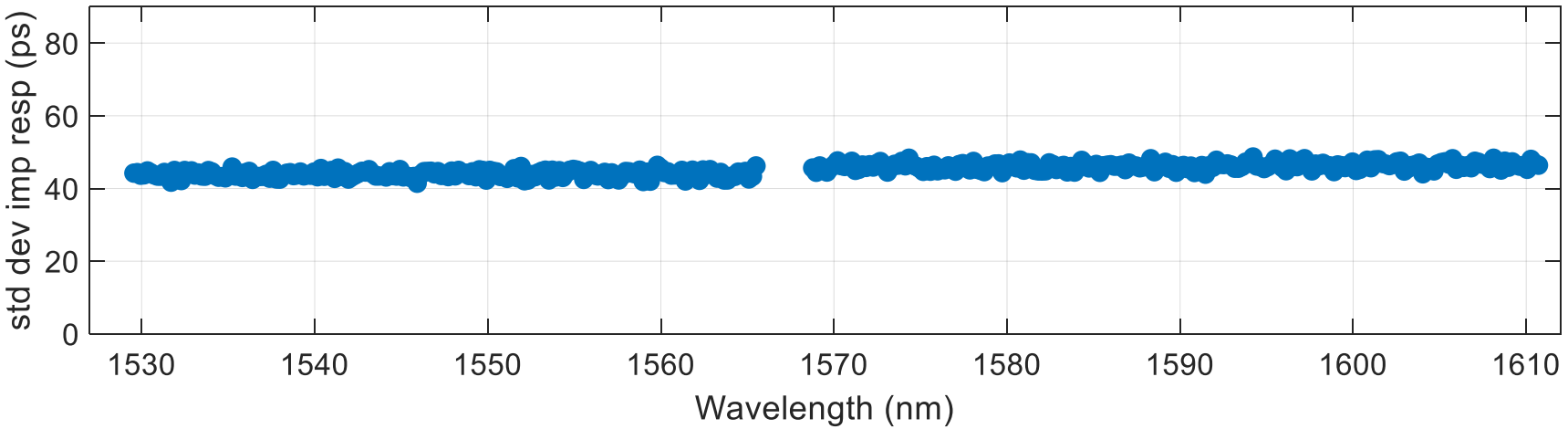
Coupling characteristics



Impulse response



Wavelength dependence of Impulse Response duration



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Thank you!

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